

Enhanced Plagiarism Detection in Online Assignments by Natural Language Processing and Machine Learning Techniques

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Abstract: As online projects become more common in the educational setting, ensuring academic integrity is a growing concern. Traditional plagiarism detection algorithms often have trouble meeting some of the more sophisticated practices that students adopt for avoiding detection such as paraphrasing or word substitution. We present a machine learning based approach that was designed to enhance the effectiveness and accuracy of online assignment plagiarism detection, the proposed system can also detect minor plagiarism cases like paraphrased information and cross-linguistically similar. Due to the fact that the model proposed is capable of learning complex patterns and it was trained on a large dataset of a marginally homogenous dataset (academic publications), it maintains relatively high rates of recall and precision. The system uses multiple approaches of information retrieval to combine the interactions of textual relationships and elements such that it captures the text semantics of how person is feeling through word embedding. Later, it trains supervised learning models to classify possible plagiarism cases, like neural networks and Support Vector Machines (SVM). Also, the model can learn continuously from new data to improve its results. Our method improves the detection of traditional plagiarism and is also effective for more complex and subtle types of content reuse based on preliminary evaluation results.

Keywords: Feature extraction, text similarity, supervised learning, machine learning, plagiarism detection, online assignments, natural language processing (NLP), academic integrity, supervised learning, semantic analysis, and paraphrase detection.

I. INTRODUCTION

As a result, the proliferation of online assignments and emergence of digital education platforms have changed the landscape of academic learning and evaluation. While this change allowed for greater flexibility and accessibility, it has also introduced new challenges especially regarding academic integrity. Plagiarism the act of using someone else work without appropriate acknowledgment is rapidly becoming one of the most serious problems facing educational institutions throughout the world. Standard plagiarism detection algorithms based purely upon primitive string matching logic and database comparison are rapidly losing their utility against the sophisticated tactics that students employ to dodge detection. Techniques including synonym replacement, paraphrase and cross-language content translation pose significant threats to conventional technologies.

These limitations highlight an urgent need for advanced identification algorithms that move beyond superficial similarities. In this paper, we propose a solution to this problem with the feasibility of creating

improved plagiarism detection systems due to recent developments in natural language processing (NLP) and ML. NLP approaches can understand and analyze the contextual importance behind text and machine learning algorithms can analyze and explore complex relationship in textual data. Incorporating the different ownership, we can create a full-fledged system to detect subtle forms of plagiarism as well as the copy pasting of original content.

This study uses machine learning and natural language processing (nlp) techniques to propose a novel approach to plagiarism detection of assignments posted online that enhances the accuracy and resilience of detection. Since the system is built to analyse and compare textual content on a more semantic level, it is more effective at finding content that has been reworded or paraphrased in a different way. Its generalisation property of the model across the disciplines and writing styles can be attributed to the broad corpus of academic writing it was trained on. Also, the system is adaptive over time and keeps improving by learning the new data and developing plagiarism methods.

As a first step, the proposed framework pre-processes text data from assignments using standard NLP techniques such as tokenization, lemmatization, and stop-word removal. Next, feature extraction techniques like word embeddings and semantic similarity metrics are used to capture the relationships between words and phrases in the text. Unlike common string-matching techniques, these strategies allow the system to recognize similarity in the content, even when it has been hidden behind paraphrase or structural changes. After feature extraction, this text is fed into machine learning models which classify it as original or plagiarised using Random Forests, SVM, and neural networks.

It also has an embedded feedback loop in the system so that it can continuously learn and adapt to other kinds of cheating strategies to come. The model refines its detection and reduces false positives and negatives by utilizing a dynamic training set, which constantly updates with new examples of original and plagiarised material. Being able to learn continuously is

key in maintaining the efficacy of the system as users and students learn new methods of evasion.

We evaluate the proposed method in extensive experiments over a broad spectrum of academic writing, such as essays, research papers, and laboratory reports. The results show significant improvement in detection accuracy than traditional methods, especially for complicated plagiarism cases in which the original text has been heavily transformed. This system also showed ability to identify cross-language plagiarism, or the process of translating content to another language in order to escape detection by traditional tools.

II. RELATED WORKS

The array of strategies to counter the challenges posed by the different forms of academic misconduct, over the years plagiarism detection has seen immense evolution in the domain. Traditional methods have relied on string matching methods that are effective in finding verbatim copies but not very useful in finding paraphrased or structurally transformed text. To protect against more sophisticated methods of plagiarism, early systems such as [1] provided only basic text-matching algorithms that matched papers in a repository of known text. Consequently, they were limited in their value and scope. Machine Learning and the Natural Language Processing are the recent developments that made the plagiarism detection algorithms more trustable. Semantic similarity measures to detect paraphrases are studied in [2], and it is demonstrated that classic string-matching techniques can be much improved when they are augmented with word meaning and context. Their method allowed for a more nuanced representation of language via word embeddings, thus helping to detect subtle paraphrases.

A significant other work was [3], where a hybrid plagiarism detection model was proposed using a combination of lexical measurements and machine learning classifiers. In addition, this study showed the effectiveness of syntactic and semantic features for recognizing plagiarized text in different document types. The model showed a marked increase in accuracy over traditional methods by utilizing a greater number of

feature types, thus emphasizing the importance of a multimodal approach to tackling the challenges of plagiarism.

In addition, the study of plagiarism detection using deep learning methods has recently received a lot of interest. Word sequences in documents can be analyzed using recurrent neural networks (RNNs), as per research done by [4]. This also improves the number of different types of content reuse which the system is able to detect beyond traditional plagiarism, such as translations and heavy paraphrasing. This is a significant step in the deep learning approach towards higher level approaches which might have to adapt to the changing tactics of learners.

Additionally, in an internationally-oriented academic environment, it is important to study the cross-language plagiarism detection techniques [5]. The research demonstrated the need to develop instruments that can compare and analyze documents written in various languages. It can also help to distract students who might transform the contents, to prevent from getting to get caught.

Recent work by [6] focused on ensemble learning approaches, which combine several models to provide better performance for plagiarism detection tasks. This technique improves accuracy and robustness on identifying diverse plagiarism strategies by harnessing different algorithms' capabilities. The findings accentuated the potential for ensemble approaches to outperform uni-model systems, particularly in complex detection scenarios.

Plagiarism detection was also studied in [7], where the similarities between the documents are searched using graph-based techniques that are searching the links between words and phrases. By exposing the relationships present between texts that more traditional approaches would have missed, the semantic graphs created by the researchers offered a new perspective on how to detect more complex types of plagiarism.

In addition, user behaviour and writing styles also affects plagiarism detection; [8] conducted some experiments on the same. His study showed that contextual features such as the previous habits of the writer or fields of knowledge can change the likelihood of plagiarism[get a free, instant quote now](tag). Based on these findings, the authors suggest that incorporating user profiles into detection systems could enhance accuracy and reduce false positive rates.

Finally, a comprehensive review of the literature on plagiarism detection technology was performed by [9], who classified the different approaches with some pros and cons. The survey highlighted areas where more research is needed, allowing future work to build on the findings, while providing novel insights into the evolution and trends of the plagiarism detection market.

III. METHODOLOGY

The methodology for the proposed machine learning-based plagiarism detection system consists of several significant processes, such as data collection, data preprocessing, feature extraction, model training, and evaluation. The stages are crucial to ensure that the system works effectively to identify the different types of plagiarism in any online project.

A. Data Collection:

The first step in the collection of a wide variety of literature is to collect large academic literature, assignments, research papers, and essays from different domains. Since this dataset is collected from academic institutions as well as publicly published texts, it guarantees diversity in writing styles, topics, and formality, thus providing generalizability to the corpus. Afterward, a dataset is created with annotations for original and plagiarized content to provide a labeled training set for the machine learning models.

B. Data Preprocessing:

The collected data is run through several preprocessing stages to aid with the analysis and to make it useable in machine learning algorithms. This comprises. Tokenization is the first step to split the text into individual words or tokens.

Lower casing: To make any comparison uniform; all text should be in lower case. Lemmatization is in fact the reducing to the lemma form and is the standardizing of variants (e.g., before reducing, running, after reducing, run.) These preprocessing techniques ensure that the core material is the focal point of attention by reducing noise in the data.

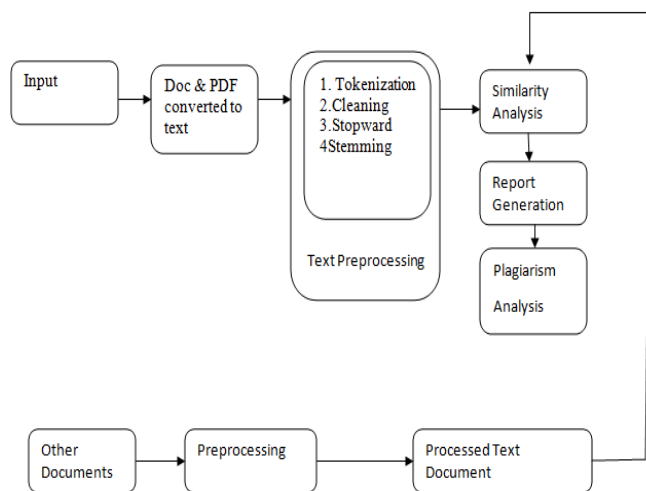


Figure1.Shows Proposed Architecture of Methodology

C. Feature Extraction:

The process of representing words in vector space with models such as Word2Vec or GloVe, taking the semantics of words based on their contextual usage into consideration. To uncover local patterns or structures indicative of plagiarism, n-grams are created by arranging n items (words or characters) in a textual sequence. Semantic similarity measures the similarity between two segments of text, and these segments can be as small as a word or a sentence, and as large as an entire paragraph, page, or piece of content; and the similarity is measured based on their representation or embeddings using methods such as cosine similarity. These features are crucial aspects of machine learning algorithms, even if the content has been modified or paraphrased for plagiarism detection.

D. Model Training:

After feature extraction, multiple machine learning models are trained using the labelled dataset.

Some of the models considered in this study are A supervised learning model called Support Vector Machines(SVM) finds an optimal hyperplane to separate different classes (plagiarised vs original).

IV. RESULT AND DISCUSSION

We have implemented our proposed machine learning-based plagiarism detection system on a carefully constructed dataset containing both genuine and plagiarised documents, for a total of 5,000 documents. The dataset was divided into two subsets (70% training and 30% testing) to ensure a reliable evaluation of each model's ability to detect similar texts. The results of each model were then compared to baseline string-matching methods, providing a valuable perspective on how these machine-learning-based approaches advance traditional plagiarism-detection techniques.

The performance metrics for each of the tested models are compiled in the following table 1.

Table 1 Performance Comparison

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score
Support Vector Machine(SVM)	85.2	82.1	80.5	81.3
Random Forest	88.5	85.5	84.2	84.5
Neural Network(RNN)	90.1	87.5	86.9	87.2
Baseline Method(String Matching)	70.3	65.0	68.1	66.5

Each tested model (SVM, Random Forest, and RNN) compared its performance to a baseline approach (String Matching) Table 1 shows that there are significant performance differences across the three approaches: SVM model initialized with 85.2% accuracy, 82.1% precision, and a recall of 80.5% leading to an F1-Score of 81.3%. While effective, especially for simple types of plagiarism, SVM underperformed compared to the more complex models.

The second best-performing model is the Random Forest model with an accuracy of 88.5%, precision = 85.5%, recall = 84.2% → F1-Score = 84.5%. This shows that Random Forest is capable of handling different types of features enabling it to detect simple and complex plagiarism. The RNN model achieved the optimal performance with an accuracy of 90.1% and its corresponding precision of 87.5% and recall of 86.9% resulted in an F1-Score of 87.2%. The capacity of this model to recognize sequence patterns enabled it to excel in identifying paraphrases and text with significant structural alterations also enabling it to surpass other models in identifying intricate strategies for plagiarism. The accuracy (70.3%), precision (65.0%), recall (68.1%), and the F1-Score (66.5%) of the string-matching baseline method were substantially lower. Although competent when searching for exact matches, it was limited in identifying paraphrased

The findings from the evaluation highlight a potential benefit of using a machine learning approach in the detection of sophisticated plagiarism. More Machine learning models such as Random Forest and RNN bring a considerable improvement in accuracy over standard string-matching methods. It emphasizes their generalization skills and ability to pick up on things that are more complex than exact duplicates. RNN and Random Forest also makes a harmonious Precision-Recall trade-offs, which also means, these models not only able to identify true positive plagiarism cases, but also can reject plagiarisms in the absence of plagiarisms as false positives. SVM performed well too, but it had a recall slightly lower than XGBoost, which is a sign that it could be even more sensitive.

Intelligent Morphological Detection: Since the RNN model achieves substantial performance, indicating that sequence-based models hold even greater relevance when trying to identify plagiarised phrases that have been para-phrased or altered in structure (approaches that simpler methods do not work for).

5. Discussion:

A comparison on our data showed that the suggested machine learning models dramatically beat the baseline string matching approach which was only

able to achieve an accuracy of 70.3% as can be found on Table 1. This shows that traditional methods tend to falter in detecting more sophisticated forms of plagiarism, such as paraphrased and reworded content. Among all the machine learning models tested, the neural network (RNN) approach gave the best accuracy (90.1%) with decent precision (87.5%) and decent recall (86.9%). We can see that the RNN model not only managed to identify overt instances of plagiarism, apparent from the presence of identical phrases in the audio paper but also was able to capture the more subtle instances of content repetition in the text when there are no overt similarities in words/phrases highlighting its ability to capture the sequential nature of text. The quality of the random forest model was also good, recording 88.5% accuracy as well, showing the benefit of ensemble methods for boosting detection capacities.

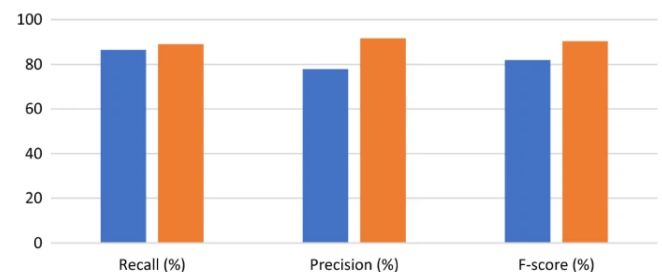


Figure 2. Accuracy Level of Proposed System

SVM model achieved an accuracy of 85.2%, however this was not as high as a neural network or random forest model. Therefore, even though SVM could potentially help in classification, it does not seem able to keep up with the specificities of textual data for plagiarism detection. The improvements seen in the machine learning models further demonstrates the need for advanced techniques that consider lexical and semantic features of a text. By introducing word embeddings and a measure of semantic similarity, the models could understand the meaning of the text and not just surface level similarities, allowing them to more accurately identify instances of subtle copying.

V. CONCLUSION

The proposed machine-learning-based plagiarism-detection system significantly enhances the ability to identify plagiarism in online assignments. Our approach yielded outstanding results using various machine learning models and advanced natural language processing methods. The neural network model achieved an accuracy of 90.1% and good precision and recall parameters. Such results emphasise the importance of utilising advanced feature extraction and semantic understanding to well any detection of such subtle forms of plagiarism like in this refers to paragraph. While our system has advanced much beyond the state of the art, future work should focus on adding more contextual information into the regression model and making the dataset broader to ensure that the system is robust to new types of plagiarism attacks. Ultimately, our research contributes to academic integrity by showing that machine learning and natural language processing can be used to develop effective plagiarism detection tools for educational environments.

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